ELITE IGCSE ACADEMY

Private Past Paper Solutions

May 2024 P1HR Worked Solutions

May 2024 | P1HR

27 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

PRIVATE

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PAST PAPER QUESTION Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

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1 Here are six cards.

Five of the cards have a number written on them.

16	15	3	2	9	

Work out the number that should be written on the last card so that the mean of the six numbers will be 11

....

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The total of the 6 cards is

$$6 \times 11 = 66$$

The known total is

$$16 + 15 + 3 + 2 + 9 = 45$$

$$x = 66 - 45 = 21$$

FINAL ANSWER

$x = 21$.

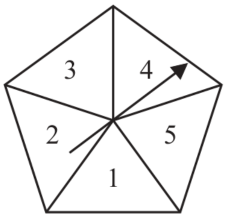
PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

2 Here is a biased spinner.



The table gives information about the probability that, when the spinner is spun once, it will land on each number.

Number	1	2	3	4	5
Probability	$2x$	0.27	0.04	x	0.12

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

....

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

Let the probability of landing on 4 be x. Then the probability of landing on 1 is $2x$.

$$2x + 0.27 + 0.04 + x + 0.12 = 1$$

$$3x = 0.57$$

$$x = 0.19$$

$$P(\text{odd}) = 0.38 + 0.04 + 0.12 = 0.54$$

Expected odd numbers:

$$400 \times 0.54 = 216$$

FINAL ANSWER

0.19, 0.54, 216.

PAST PAPER QUESTION Q#: 3 Marks: 5

TOPIC: **Percentages**

May 2024 P1HR - May 2024 | P1HR

3 Norberto sells white loaves of bread and brown loaves of bread.

He sells a total of 200 loaves such that

the number of white loaves sold : the number of brown loaves sold = 3 : 2

Norberto sells the white loaves for £1.50 each.
He sells the brown loaves for £1.75 each.

40% of the price of a white loaf is profit.
60% of the price of a brown loaf is profit.

Work out Norberto's total profit when he sells all 200 loaves.

£..............................

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 5

TOPIC: **Percentages**

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WORKED SOLUTION

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Topic check. Percentages. The tag is correct, with ratio used to split the loaves.

Method

The ratio parts are

$$3 + 2 = 5$$

White loaves:

$$200 \times \frac{3}{5} = 120$$

Brown loaves:

$$200 \times \frac{2}{5} = 80$$

Profit on each white loaf:

$$40\% \times 1.50 = 0.60$$

Profit on each brown loaf:

$$60\% \times 1.75 = 1.05$$

Total profit:

$$120 \times 0.60 + 80 \times 1.05 = 72 + 84 = 156$$

FINAL ANSWER

156 pounds

PAST PAPER QUESTION

Q#: 4 Marks: 3

TOPIC: **Algebraic Proof**

May 2024 P1HR - May 2024 | P1HR

4 Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Algebraic Proof**

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Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$2\frac{1}{3} = \frac{7}{3}, \qquad 5\frac{1}{4} = \frac{21}{4}$$

$$2\frac{1}{3} \div 5\frac{1}{4} = \frac{7}{3} \div \frac{21}{4}$$

$$= \frac{7}{3} \times \frac{4}{21} = \frac{28}{63} = \frac{4}{9}$$

FINAL ANSWER

$\frac{4}{9}$.

PAST PAPER QUESTION

Q#: 5 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2024 P1HR - May 2024 | P1HR

5 Slavomir invests 5200 euros in a savings account for 4 years.
He gets 2.5% per year compound interest.

Work out how much money Slavomir will have in the savings account at the end of 4 years.
Give your answer correct to the nearest euro.

.... euros

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

The multiplier for 2.5% compound interest is

$$1.025$$

After 4 years,

$$5200(1.025)^4 = 5739.827\ldots$$

Correct to the nearest euro,

$$5739.827\ldots \approx 5740$$

FINAL ANSWER

5740 euros

PAST PAPER QUESTION

Q#: 6 Marks: 4

TOPIC: **Standard & Compound Units**

May 2024 P1HR - May 2024 | P1HR

6 The diagram shows a solid wooden cylinder.

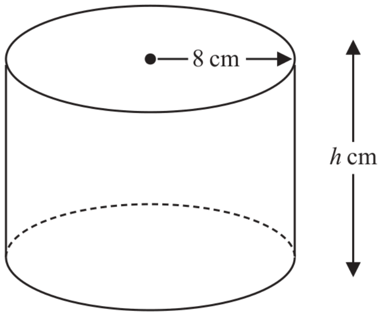

Diagram **NOT** accurately drawn

The cylinder has radius 8 cm and height h cm.
The volume of the cylinder is 1208 cm^3

(a) Work out the value of h
Give your answer correct to the nearest whole number.

h = ...

(2)

The density of the wood is 1.25 g/cm^3

(b) Work out the mass of the cylinder.
Give your answer in kilograms.

.... kilograms

(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Standard & Compound Units**

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Topic check. Standard & Compound Units. The tag is correct.

Solution

For the cylinder,

$$V = \pi r^2 h$$

$$1208 = \pi(8)^2 h$$

$$h = \frac{1208}{64\pi} = 6.006\ldots$$

So, to the nearest whole number,

$$h = 6$$

For the mass, use the given volume:

$$\text{mass} = \text{density} \times \text{volume}$$

$$1.25 \times 1208 = 1510 \text{ g}$$

$$1510 \text{ g} = 1.51 \text{ kg}$$

FINAL ANSWER

$h = 6$ cm, and the mass is 1.51 kg.

PAST PAPER QUESTION

Q#: 7 Marks: 9

TOPIC: **Factorising**

May 2024 P1HR - May 2024 | P1HR

7 (a) Simplify $g^9 \div g^2$

....

(1)

(b) Expand $5k^2(k^3 + 4)$

....

(2)

(c) (i) Factorise $x^2 - 2x - 63$

....

(2)

(ii) Hence, solve $x^2 - 2x - 63 = 0$

....

(1)

(d) Solve the inequality $7 - 2y < 3y - 12$

....

(3)

(Total for Question 7 is 9 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 9

TOPIC: **Factorising**

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WORKED SOLUTION

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Topic check. Corrected to factorising. The original tag was expanding brackets, but the quadratic factorising section is the strongest classification in this mixed algebra question.

Solution

(a) $g^9 \div g^2 = g^7$.
(b) $5k^2(k^3 + 4) = 5k^5 + 20k^2$.
(c)(i)

$$x^2 - 2x - 63 = (x - 9)(x + 7)$$

(c)(ii) $x = 9$ or $x = -7$.
(d)

$$7 - 2y < 3y - 12$$

$$19 < 5y$$

FINAL ANSWER

(a) g^7, (b) $5k^5 + 20k^2$, (c) $(x - 9)(x + 7)$, $x = 9$ or $x = -7$, (d) $y > \frac{19}{5}$

PAST PAPER QUESTION Q#: 8 Marks: 6

TOPIC: **Area & Perimeter**

May 2024 P1HR - May 2024 | P1HR

8 The diagram shows a trapezium, *ABCD*

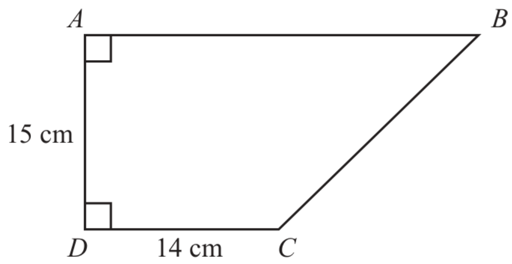

Diagram **NOT** accurately drawn

DAB and *ADC* are right angles.

$AD = 15$ cm $\qquad DC = 14$ cm

The area of the trapezium is 360 cm²

Work out the perimeter of the trapezium.

.... cm

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 6

TOPIC: **Area & Perimeter**

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WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

Use the trapezium area formula:

$$360 = \frac{1}{2}(AB + 14)(15)$$

$$AB + 14 = 48$$

$$AB = 34$$

Triangle ABC is right-angled with horizontal length

$$34 - 14 = 20$$

So

$$BC = \sqrt{20^2 + 15^2} = 25$$

Perimeter:

$$15 + 14 + 25 + 34 = 88$$

FINAL ANSWER

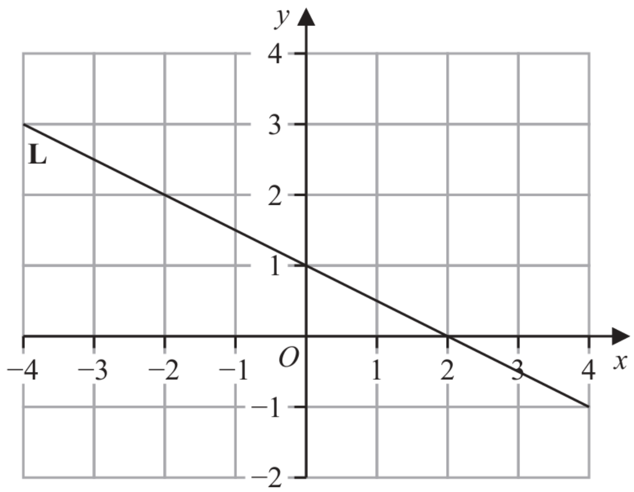
88 cm.

PAST PAPER QUESTION Q#: 9 Marks: 3

TOPIC: **Linear Graphs (y = mx + c)**

May 2024 P1HR - May 2024 | P1HR

9 Line **L** is drawn on the grid.



Find an equation for **L**
Give your answer in the form $y = mx + c$

....

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Linear Graphs (y = mx + c)**

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WORKED SOLUTION

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Topic check. Corrected to linear graphs. The question asks for an equation of a drawn line.

Solution

From the graph, the line crosses the y-axis at 1, so

$$c = 1$$

It also crosses the x-axis at 2, so using $(0, 1)$ and $(2, 0)$,

$$m = \frac{0-1}{2-0} = -\frac{1}{2}$$

Therefore

$$y = -\frac{1}{2}x + 1$$

FINAL ANSWER

$y = -\frac{1}{2}x + 1$.

PAST PAPER QUESTION

Q#: 10 Marks: 2

Topic: **Statistics Toolkit**

May 2024 P1HR - May 2024 | P1HR

10 Here are the numbers of goals scored by a hockey team in its 11 games this season.

0 1 2 2 3 4 4 6 7 9 11

Work out the interquartile range of the numbers of goals.

....

(Total for Question 10 is 2 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 2

TOPIC: **Statistics Toolkit**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The goals in order are

$$0,\ 1,\ 2,\ 2,\ 3,\ 4,\ 4,\ 6,\ 7,\ 9,\ 11$$

$$Q_1 = 2, \qquad Q_3 = 7$$

$$\text{IQR} = 7 - 2 = 5$$

FINAL ANSWER

5.

PAST PAPER QUESTION

Q#: 11 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2024 P1HR - May 2024 | P1HR

11 $A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

(a) Write down the highest common factor (HCF) of $5A$ and $2B$
Give your answer as a product of prime factors.

....

(2)

$A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

(b) Work out the value of $(AB)^2$
Give your answer as a product of prime factors.

....

(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$A = 2^5 \times 5 \times 7^2$$

$$B = 2^3 \times 5^3 \times 7^4$$

For part (a),

$$5A = 2^5 \times 5^2 \times 7^2$$

$$2B = 2^4 \times 5^3 \times 7^4$$

The HCF takes the smaller powers:

$$\text{HCF} = 2^4 \times 5^2 \times 7^2$$

For part (b),

$$AB = 2^{5+3} \times 5^{1+3} \times 7^{2+4}$$

$$AB = 2^8 \times 5^4 \times 7^6$$

Therefore

$$(AB)^2 = 2^{16} \times 5^8 \times 7^{12}$$

FINAL ANSWER

(a) $2^4 \times 5^2 \times 7^2$, (b) $2^{16} \times 5^8 \times 7^{12}$

PAST PAPER QUESTION Q#: 12 Marks: 4

TOPIC: **Simultaneous Equations**

May 2024 P1HR - May 2024 | P1HR

12 Solve the simultaneous equations

$$4x + 3y = 9.6$$
$$6x + 5y = 16.8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Simultaneous Equations**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Simultaneous equations. The tag is correct.

Solution

$$4x + 3y = 9.6$$

$$6x + 5y = 16.8$$

Multiply the first equation by 5:

$$20x + 15y = 48$$

Multiply the second equation by 3:

$$18x + 15y = 50.4$$

Subtract:

$$2x = -2.4$$

$$x = -1.2$$

Substitute into $4x + 3y = 9.6$:

$$4(-1.2) + 3y = 9.6$$

$$3y = 14.4$$

$$y = 4.8$$

FINAL ANSWER

$x = -1.2$, $y = 4.8$.

PAST PAPER QUESTION

Topic: **Circle Theorems**

May 2024 P1HR - May 2024 | P1HR

Q#: 13 **Marks: 5**

13

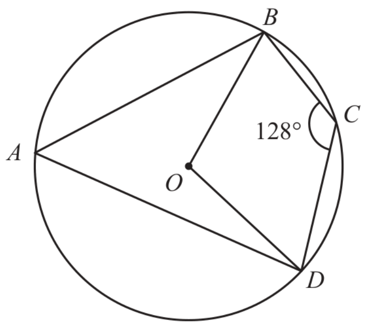

Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle, centre *O*

Angle *BCD* = 128°

Work out the size of angle *OBD*
Give a reason for each stage of your working.

angle *OBD* = ... °

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 5

TOPIC: **Circle Theorems**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Circle Theorems. The tag is correct.

Solution

Opposite angles in cyclic quadrilateral $ABCD$ add to $180°$:

$$\angle BAD = 180° - 128° = 52°$$

The angle at the centre is twice the angle at the circumference standing on the same arc BD:

$$\angle BOD = 2 \times 52° = 104°$$

Since $OB = OD$, triangle OBD is isosceles:

$$\angle OBD = \frac{180° - 104°}{2} = 38°$$

FINAL ANSWER

$38°$.

PAST PAPER QUESTION

Q#: 14 Marks: 6

TOPIC: **Expanding Brackets**

May 2024 P1HR - May 2024 | P1HR

14 (a) Expand and simplify $(3x+1)(2-x)(4+x)$

....

(3)

(b) Simplify fully $\left(\dfrac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

....

(3)

(Total for Question 14 is 6 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 6

TOPIC: **Expanding Brackets**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Expanding brackets. The tag is correct.

Solution

(a)

$$(2-x)(4+x) = 8 - 2x - x^2$$

$$(3x+1)(8-2x-x^2) = -3x^3 - 7x^2 + 22x + 8$$

(b)

$$\left(\frac{a^3b}{a^9b^5}\right)^{-1/2} = \left(\frac{1}{a^6b^4}\right)^{-1/2} = a^3b^2$$

FINAL ANSWER

(a) $-3x^3 - 7x^2 + 22x + 8$, (b) a^3b^2

PAST PAPER QUESTION

Q#: 15 **Marks: 3**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2024 P1HR - May 2024 | P1HR

15 The diagram shows isosceles triangle *EFG*

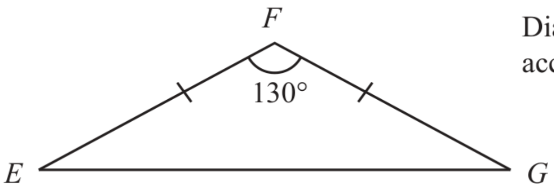

Diagram **NOT** accurately drawn

$EF = GF$
Angle $EFG = 130°$
The area of triangle EFG is 74 cm^2

Work out the length of EF
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Sine, Cosine Rule & Area of Triangles. The tag is correct.

Solution

Let

$$EF = FG = x$$

Use the area formula:

$$74 = \frac{1}{2}x^2 \sin 130^\circ$$

$$x^2 = \frac{148}{\sin 130^\circ}$$

$$x = 13.899\ldots$$

FINAL ANSWER

13.9 cm.

PAST PAPER QUESTION Q#: 16 Marks: 3

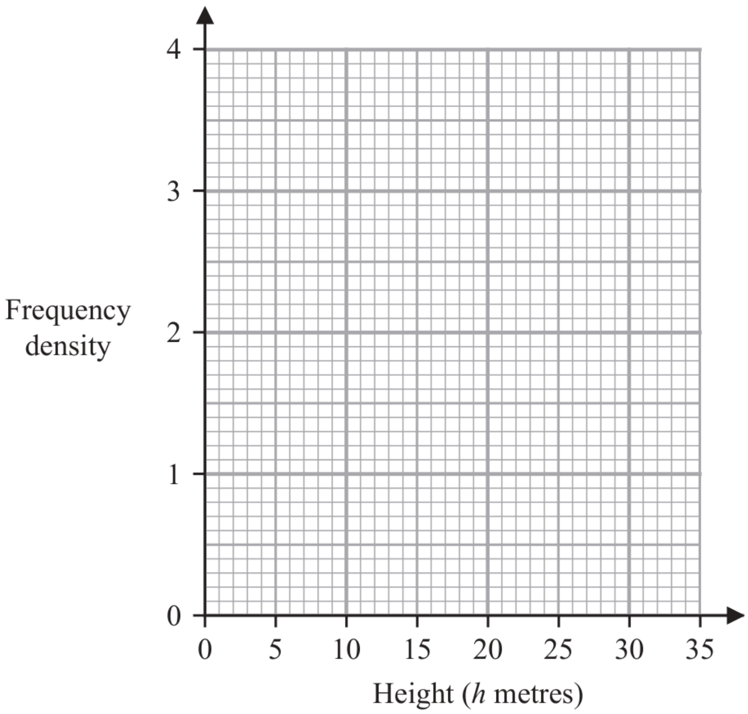
TOPIC: **Histograms**

May 2024 P1HR - May 2024 | P1HR

16 The table gives information about the heights, in metres, of the trees in a park.

Height (*h* metres)	**Frequency**
$0 < h \leqslant 2$	5
$2 < h \leqslant 5$	12
$5 < h \leqslant 10$	18
$10 < h \leqslant 20$	14
$20 < h \leqslant 35$	9

On the grid, draw a histogram for this information.



(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Histograms**

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Topic check. Histograms. The tag is correct.

Solution

For a histogram,

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

The bar heights are:

$$0 < h \leq 2 : \frac{5}{2} = 2.5$$

$$2 < h \leq 5 : \frac{12}{3} = 4$$

$$5 < h \leq 10 : \frac{18}{5} = 3.6$$

$$10 < h \leq 20 : \frac{14}{10} = 1.4$$

$$20 < h \leq 35 : \frac{9}{15} = 0.6$$

FINAL ANSWER

draw bars with heights $2.5, 4, 3.6, 1.4, 0.6$.

PAST PAPER QUESTION Q#: 17 Marks: 4

TOPIC: **Algebraic Roots & Indices**

May 2024 P1HR - May 2024 | P1HR

17 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

$n =$ ....

(1)

(b) Express $\dfrac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

....

(3)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Algebraic Roots & Indices**

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Topic check. Algebraic roots and indices. The tag is correct.

Solution

(a)

$$\left(\sqrt[4]{k^{12}}\right)^5 = (k^3)^5 = k^{15}$$

so $n = 15$.

(b)

$$\frac{7}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{7(2+\sqrt{3})}{4-3} = 14 + 7\sqrt{3}$$

$$7\sqrt{3} = \sqrt{49 \times 3} = \sqrt{147}$$

FINAL ANSWER

(a) 15, (b) $\sqrt{147} + 14$

PAST PAPER QUESTION

Q#: 18 Marks: 4

Topic: **Area & Volume of Similar Shapes**

May 2024 P1HR - May 2024 | P1HR

18 The diagram shows two similar vases, **A** and **B**

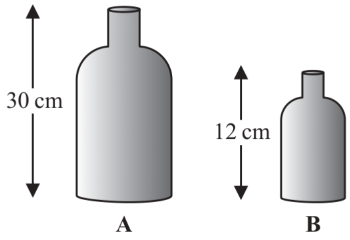

Diagram **NOT** accurately drawn

The height of vase **A** is 30 cm
The height of vase **B** is 12 cm

Given that

surface area of vase **A** – surface area of vase **B** = 178.5 cm²

find the surface area of vase **A**

.... cm²

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

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Topic check. Area & Volume of Similar Shapes. The tag is correct.

Solution

The linear scale factor from vase B to vase A is

$$\frac{30}{12} = 2.5$$

So the surface area scale factor is

$$2.5^2 = 6.25$$

Let the surface area of vase B be S.

$$6.25S - S = 178.5$$

$$5.25S = 178.5$$

$$S = 34$$

Therefore the surface area of vase A is

$$6.25 \times 34 = 212.5$$

FINAL ANSWER

212.5 cm^2.

PAST PAPER QUESTION

Q#: 19 Marks: 5

TOPIC: **Differentiation**

May 2024 P1HR - May 2024 | P1HR

19 A curve C has equation $y = x^3 - 8x^2 - 12x + 5$

Curve C has exactly two stationary points, one at point A and one at point B such that

$$x \text{ coordinate of point } A > x \text{ coordinate of point } B$$

Find the coordinates of point A
Show clear algebraic working.

(.............................. , ...)

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Differentiation**

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WORKED SOLUTION

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Topic check. Correct. This is finding stationary points using differentiation.

Solution

$$y = x^3 - 8x^2 - 12x + 5$$

$$\frac{dy}{dx} = 3x^2 - 16x - 12$$

Stationary points have

$$3x^2 - 16x - 12 = 0$$

$$x = \frac{16 \pm \sqrt{(-16)^2 - 4(3)(-12)}}{6}$$

$$x = \frac{16 \pm 20}{6}$$

$$x = 6 \quad \text{or} \quad x = -\frac{2}{3}$$

Point A has the greater x-coordinate, so $x = 6$.

$$y = 6^3 - 8(6^2) - 12(6) + 5$$

$$y = 216 - 288 - 72 + 5 = -139$$

FINAL ANSWER

$A = (6, -139)$.

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Completing the Square**

May 2024 P1HR - May 2024 | P1HR

20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a, b and c are numbers to be found.

....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(.................... , ...)

(2)

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 5

TOPIC: **Completing the Square**

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WORKED SOLUTION

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Topic check. Moved from Coordinate Geometry to Completing the Square.

Method

For part (a),

$$2x^2 - 11x + 9$$

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

For part (b),

$$y = 2(x-3)^2 - 11(x-3) + 9$$

Let

$$u = x - 3$$

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

The minimum value of y is

$$-\frac{49}{8}$$

FINAL ANSWER

$2\left(x-\frac{11}{4}\right)^2-\frac{49}{8}$, $P=\left(\frac{23}{4},-\frac{49}{8}\right)$

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Completing the Square**

May 2024 P1HR - May 2024 | P1HR

20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a, b and c are numbers to be found.

....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(.............................. , ...)

(2)

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Completing the Square**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Completing the Square.

Method

For part (a),

$$2x^2 - 11x + 9$$

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

For part (b),

$$y = 2(x-3)^2 - 11(x-3) + 9$$

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$$x = \frac{23}{4}$$

The minimum value of y is

$$-\frac{49}{8}$$

FINAL ANSWER

$2\left(x-\frac{11}{4}\right)^2-\frac{49}{8}$, $P=\left(\frac{23}{4},-\frac{49}{8}\right)$

PAST PAPER QUESTION

Q#: 21 **Marks: 5**

TOPIC: **Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

21 There are 25 counters in a bag such that

6 counters are blue
x counters are orange, where $x > 9$
the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.
Show clear algebraic working.

....

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 5

TOPIC: **Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

There are 25 counters.
Blue counters:

$$6$$

Orange counters:

$$x$$

Pink counters:

$$25 - 6 - x = 19 - x$$

The probability of taking one orange and one pink in either order is

$$\frac{x}{25} \cdot \frac{19-x}{24} + \frac{19-x}{25} \cdot \frac{x}{24}$$

$$= \frac{2x(19-x)}{25 \cdot 24}$$

This is equal to $\frac{22}{75}$, so

$$\frac{2x(19-x)}{600} = \frac{22}{75}$$

$$\frac{x(19-x)}{300} = \frac{22}{75}$$

$$x(19-x) = 88$$

$$19x - x^2 = 88$$

$$x^2 - 19x + 88 = 0$$

$$(x-8)(x-11) = 0$$

Since $x > 9$,

$$x = 11$$

So the number of pink counters is

$$19 - 11 = 8$$

The probability of taking 2 pink counters is

$$\frac{8}{25} \cdot \frac{7}{24} = \frac{56}{600}$$
$$= \frac{7}{75}$$

FINAL ANSWER

$\frac{7}{75}$

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

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Show clear algebraic working.

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(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Topic: **Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

Q#: 21 Marks: 5

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

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$$= \frac{7}{75}$$

FINAL ANSWER

$\frac{7}{75}$

PAST PAPER QUESTION

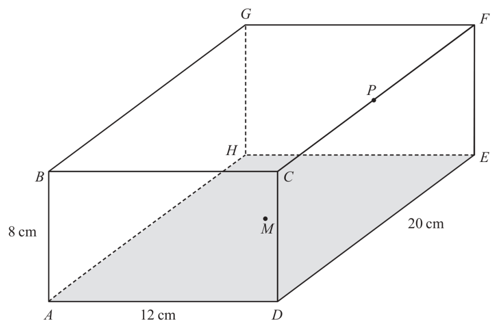
Q#: 22 **Marks: 6**

Topic: **3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

22 The diagram shows a cuboid *ABCDEFGH* with horizontal base *ADEH*

Diagram **NOT** accurately drawn



AB = 8 cm *AD* = 12 cm *DE* = 20 cm

M is the midpoint of the base *ADEH* and *P* is the midpoint of the edge *CF*

Work out the size of angle *BMP*
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Use coordinates with

$$A = (0, 0, 0)$$

Since

$$AD = 12, \qquad DE = 20, \qquad AB = 8$$

take

$$B = (0, 0, 8)$$

The centre of the rectangular base $ADEH$ is

$$M = (6, 10, 0)$$

Point P is the midpoint of CF.

$$C = (12, 0, 8), \qquad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

FINAL ANSWER

$78.6°$

PAST PAPER QUESTION

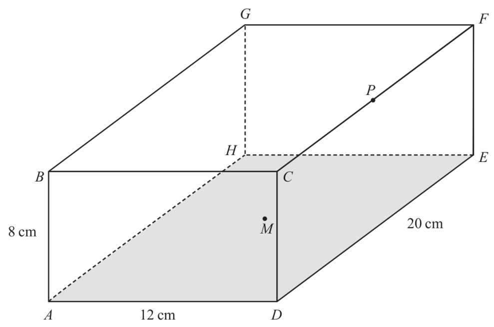
Q#: 22 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

22 The diagram shows a cuboid *ABCDEFGH* with horizontal base *ADEH*

Diagram **NOT** accurately drawn



$AB = 8$ cm $\quad AD = 12$ cm $\quad DE = 20$ cm

M is the midpoint of the base *ADEH* and *P* is the midpoint of the edge *CF*

Work out the size of angle *BMP*
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Use coordinates with

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Since

$$AD = 12, \qquad DE = 20, \qquad AB = 8$$

take

$$B = (0, 0, 8)$$

The centre of the rectangular base $ADEH$ is

$$M = (6, 10, 0)$$

Point P is the midpoint of CF.

$$C = (12, 0, 8), \qquad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

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$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

FINAL ANSWER
$78.6°$

PAST PAPER QUESTION Q#: 23 Marks: 4

TOPIC: **Sequences**

May 2024 P1HR - May 2024 | P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 4

TOPIC: **Sequences**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The first three terms are

$$4x - 14, \quad x + 2, \quad 7x - 9$$

Equal differences give

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$$

$$-3x + 16 = 6x - 11$$

$$9x = 27$$

$$x = 3$$

So the sequence is

$$-2,\ 5,\ 12, \ldots$$

with common difference 7.

$$S_{40} = \frac{40}{2}(2(-2) + 39(7))$$

$$S_{40} = 20(-4 + 273) = 5380$$

FINAL ANSWER

5380

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Sequences**

May 2024 P1HR - May 2024 | P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Sequences**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The first three terms are

$$4x - 14, \quad x + 2, \quad 7x - 9$$

Equal differences give

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$$

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FINAL ANSWER

5380